

Exercise 1

$\varinjlim$ is a functor.

Exercise 2

(1) Let $x \in X$ and $[(U, s)] \in \ker \varphi_x$. Then $[(U, \varphi_U(s))] = 0$ in $\mathcal{G}_x$, i.e. there exists an open neighbourhood $x \in V \subseteq U$ s.t. $0 = \varphi_U(s)|_V = \varphi_V(s|_V)$. But φ_V is injective by assumption, so $s|_V = 0$. Hence $[(U, s)] = [(V, s|_V)] = 0$ and φ_x is injective. Since x was arbitrary, we conclude that φ is injective.

For a counterexample of the reverse implication, let $|X| = 2$ with the discrete topology, let $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$. Since all restrictions are trivial, this is clearly a presheaf. Furthermore, for any $x \in X$, $\{x\}$ is final in the directed system $U \ni x$, so $\mathcal{F}_x = \mathcal{F}(\{x\}) = 0$. Now let $\mathcal{G} = 0$ on X and consider the unique morphism $\mathcal{F} \rightarrow \mathcal{G}$ (again it is obvious that this is a morphism since all restrictions are trivial). Since all stalks are trivial, this is injective, but $\mathbb{Z} = \mathcal{F}(X) \rightarrow \mathcal{G}(X) = 0$ is not.

(2) This was proven as part of theorem 1.11.

(3) Let $x \in X$ and $[(U, t)] \in \mathcal{G}_x$. Since φ_U is surjective, we can find a preimage $s \in \mathcal{F}(U)$ of t . Then $\varphi_x([(U, s)]) = [(U, \varphi_U(s))] = [(U, t)]$, so φ_x is surjective. Since x was arbitrary, by definition φ is surjective.

For a counterexample of the converse, consider the presheaves of (1) and the unique morphism $\mathcal{G} \rightarrow \mathcal{F}$. For a counterexample involving sheaves, consider $\underline{\mathbb{Z}} \rightarrow i_{x,*}\mathbb{Z} \oplus i_{y,*}\mathbb{Z}$, $x \neq y \in X$, X a T_0 -space, where $i_{x,*}A$ denotes the skyscraper sheaf, the sheafification of the presheaf $U \mapsto A$ if $x \in U$ and $U \mapsto 0$ otherwise.

Exercise 3

(1) The restriction maps of $\ker \varphi$, $\operatorname{im} \varphi$, $\operatorname{coker} \varphi$ are restrictions/quotients of the restriction maps of $\mathcal{F}$ or $\mathcal{G}$. Hence they satisfy the same identities as the original maps.

(2) Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ an open cover. Let $s \in \ker \varphi(U)$ s.t. $s|_{U_i} = 0$ for all $i \in I$. Since $\ker \varphi \subseteq \mathcal{F}$ and $\mathcal{F}$ is a sheaf, this immediately implies $s = 0$ in $\mathcal{F}(U)$.

Now let $s_i \in \ker \varphi(U_i)$ be given, satisfying $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$. Again since $\mathcal{F}$ is a sheaf, these glue to a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$. It remains to check $s \in \ker \varphi(U)$. One computes

$$\varphi_U(s)|_{U_i} = \varphi_{U_i}(s|_{U_i}) = \varphi_{U_i}(s_i) = 0,$$

and using that $\mathcal{G}$ is a sheaf, we conclude $\varphi_U(s) = 0$.

Exercise 1

Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover of U . Let $s = s_1 \oplus s_2 \in (\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$ be such that $s|_{U_i} = 0$ for all $i \in I$. By definition of the restriction map, $0 = s|_{U_i} = s_1|_{U_i} \oplus s_2|_{U_i}$ implies $s_j|_{U_i} = 0$ for $j \in \{1, 2\}$. Since this holds for all $i \in I$, and $\mathcal{F}$ and $\mathcal{G}$ are sheaves, we conclude $s_j = 0$, and thus $s = s_1 \oplus s_2 = 0$.

Now let $s_i \oplus t_i \in \mathcal{F}(U_i)$ be given, s.t. $(s_i \oplus t_i)|_{U_i \cap U_j} = (s_j \oplus t_j)|_{U_i \cap U_j}$ for all $i, j \in I$. Again, by definition of the restriction maps, this means $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ and $t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}$, hence these glue to elements $s \in \mathcal{F}(U)$, $t \in \mathcal{G}(U)$ satisfying $s|_{U_i} = s_i$ and $t|_{U_i} = t_i$. Then $(s \oplus t)|_{U_i} = s|_{U_i} \oplus t|_{U_i} = s_i \oplus t_i$, so $s \oplus t \in (\mathcal{F} \oplus \mathcal{G})(U)$ is the desired element.

Exercise 2

(1) Let A be an abelian group, X a topological space, $x \in X$. Write $i : \{x\} \hookrightarrow X$ for the inclusion. Then

$$i_* \underline{A}(U) = \underline{A}(i^{-1}(U)) = \begin{cases} 0 & \text{if } i^{-1}(U) = \emptyset \Leftrightarrow x \notin U, \\ A & \text{otherwise} \end{cases} = {}^x \mathcal{F}(U),$$

so ${}^x \mathcal{F} = i_* \underline{A}$ (the restriction maps clearly agree), and the result follows from exercise 3.

(2) Let X be a connected T_1 -space, $x \neq y \in X$ and $0 \neq A$ an abelian group. We are looking at the natural map $\varphi : \underline{A} \rightarrow {}^x \mathcal{F} \oplus {}^y \mathcal{F}$. Clearly $\varphi_X : A \rightarrow A \oplus A$, $a \mapsto (a, a)$ is not surjective.

First let $z \notin \{x, y\}$. Then the set of open neighbourhoods U of z not containing x, y is cofinal in the set of all such neighbourhoods, so

$$({}^x \mathcal{F} \oplus {}^y \mathcal{F})_z = \varinjlim_{U \ni z} ({}^x \mathcal{F} \oplus {}^y \mathcal{F})(U) \cong \varinjlim_{U \ni z; x, y \notin U} ({}^x \mathcal{F} \oplus {}^y \mathcal{F})(U) = \varinjlim_{U \ni z; x, y \notin U} 0 = 0,$$

and the map $\underline{A}_z \rightarrow 0$ must be surjective.

Now let $z \in \{x, y\}$, wlog $z = x$. Similarly to above, we can compute φ_x as the limit over the cofinal set of open neighbourhoods U of x not containing y . But on any such U , $\varphi_U : A \rightarrow A \oplus 0$ is the identity by definition, hence also $\varphi_x = \text{id}_A : A \rightarrow A$.

Exercise 3

$f_* \mathcal{F}$ is clearly a presheaf with the restrictions from $\mathcal{F}$. Let $U \subseteq Y$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover. Let $s \in f_* \mathcal{F}(U)$ satisfy $s|_{U_i} = 0$ for all $i \in I$. Note that $f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U_i)$ is still an open cover since f is continuous. So $s \in \mathcal{F}(f^{-1}(U))$ is an element which is 0 when restricted to any open in that cover, hence $s = 0$ since $\mathcal{F}$ is a sheaf.

Now let $s_i \in f_* \mathcal{F}(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ be given. Again, since $f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U_i)$ is an open cover, these glue to an element of $\mathcal{F}(f^{-1}(U)) = f_* \mathcal{F}(U)$ satisfying $s|_{U_i} = s_i$, as desired.

Exercise 4

(1) Since $IJ \subseteq I \cap J$, by the lecture we have $V(I \cap J) \subseteq V(IJ) = V(I) \cup V(J)$. Conversely, since $I \cap J \subseteq I$, one gets $V(I) \subseteq V(I \cap J)$ and symmetrically $V(J) \subseteq V(I \cap J)$, hence also $V(I) \cup V(J) \subseteq V(I \cap J)$.

(2) We have

$$\begin{aligned} \mathfrak{p} \in V\left(\sum_{\alpha} I_{\alpha}\right) &\iff \sum_{\alpha} I_{\alpha} \subseteq \mathfrak{p} \iff I_{\alpha} \subseteq \mathfrak{p}, \forall \alpha \\ &\iff \mathfrak{p} \in V(I_{\alpha}), \forall \alpha \iff \mathfrak{p} \in \bigcap_{\alpha} I_{\alpha}. \end{aligned}$$

Exercise 5

First assume there exists $0, 1 \neq e \in A$ with $e^2 = e$. Then we claim that $\text{Spec } A = V(e) \sqcup V(1 - e)$: By exercise 4, $V(e) \cup V(1 - e) = V(e(1 - e)) = V(0) = \text{Spec } A$, and $V(e) \cap V(1 - e) = V((e) + (1 - e)) = V(1) = \emptyset$. Also $V(e) \neq \emptyset$, since e is a zero-divisor, hence not a unit. Since also $(1 - e)^2 = 1 - e$, the same argument applies for $V(1 - e)$ and the above is a non-trivial decomposition of $\text{Spec } A$ into two clopen subsets.

For the converse, first note that $\text{Spec } A$ is quasi-compact. Indeed, it suffices to consider coverings of the form $\text{Spec } A = \bigcup_i D(f_i)$. This is equivalent to $(f_i)_i = 1$, but this means 1 is generated by finitely many of the f_i .

Now suppose $\text{Spec } A = V(I) \sqcup V(J)$ is a non-trivial decomposition. By the same arguments as above, we have $\sqrt{0} \supseteq IJ$ and $1 = \sqrt{I + J} = I + J$. Since $V(I), V(J)$ are closed subsets of $\text{Spec } A$, they are quasi-compact as well. So we can write $V(J) = \text{Spec } A \setminus V(I) = \bigcup_i D(f_i)$ for finitely many i . Then $V(I) = V(f_i)_i$ and wlog we may assume that I, J are finitely generated.

Then IJ is finitely generated as well, and $IJ \subseteq \sqrt{0}$ implies $(IJ)^N = 0$ for some $N \gg 0$ (take the sum of exponents for a generating set). Raising $I + J = 1$ to the $2N - 1$ -th power, we see $I^N + J^N = 1$, so write $1 = e + (1 - e)$ with $e \in I^N$ and $1 - e \in J^N$. Then $e(1 - e) \in (IJ)^N = 0$, i.e. $e^2 = e$.

Exercise 1

Let $\text{Spec } A = Z_1 \cup Z_2$ with Z_1, Z_2 closed. Suppose wlog that $(0) \in Z_1$. Then $Z_1 \supseteq \overline{\{(0)\}} = \text{Spec } A$.

Exercise 2

(1) Last week.

(2) If $A = A_1 \times A_2$, then $e = (1, 0)$ is a nontrivial idempotent, so the claim follows from (1). Conversely, by (1) we have a nontrivial idempotent $e^2 = e \in A$. Let $\pi : A \rightarrow A/eA \times A/(1-e)A$ be the natural map. If $x \in \ker \pi$, then $x \in eA \cap (1-e)A = (e(1-e))A = 0$. Let $(a+eA, b+(1-e)A) \in A/eA \times A/(1-e)A$ be given. Then $\pi(a(1-e) + be) = (a(1-e) + eA, be + (1-e)A) = (a+eA, b+(1-e)A)$. So π is an isomorphism.

(3) Let $(A, \mathfrak{m})$ be local and assume A contains a non-trivial idempotent e . Then e is in particular a zero-divisor, hence not a unit, and the same applies to $1 - e$. Hence $e, 1 - e \in \mathfrak{m}$, but then also $1 = e + (1 - e) \in \mathfrak{m}$, contradiction.

Exercise 3

(1) The morphism $\pi : A \rightarrow S^{-1}A$ has the following universal property: For every morphism $f : A \rightarrow B$ such that $f(S) \subseteq B^\times$, there exists a unique morphism $\tilde{f} : S^{-1}A \rightarrow B$ with $f = \tilde{f} \circ \pi$.

(2) For the universal property to work, we only have to check that every element of $\{1, f, \dots, f^n, \dots\}$ is invertible in A_h . Clearly it suffices to do this for f . Since $D(h) \subseteq D(f) \iff V(f) \subseteq V(h)$, by Lemma 2.2 $h \in \sqrt{(f)}$, i.e. $h^m = af$ for some $a \in A$ and $m > 0$. Since h is invertible in A_h , so is f , and the result follows.

(3) These are exactly the restriction maps $\mathcal{O}(D(f)) \rightarrow \mathcal{O}(D(h))$.

Exercise 4

(1) We show more generally:

Lemma 1. Let $\mathcal{C}$ be a category with direct limits, $(I, \leq)$ a directed set indexing a directed system $(A_i)_{i \in I}$, $(\varphi_{ij} : A_i \rightarrow A_j)_{i < j}$ in $\mathcal{C}$. Let $\iota_i : A_i \rightarrow \varinjlim_{i \in I} A_i =: A$ be its direct limit. Let $J \subseteq I$ be a cofinal subset, that is, for every $i \in I$ there exists $j \in J$ with $i \leq j$. Then J is directed and $(A_j)_{j \in J}, (\varphi_{ij})_{i < j, i, j \in J}$ is a directed system; write $\iota'_j : A_j \rightarrow \varinjlim_{j \in J} A_j =: A'$ for the direct limit of this sub-directed system. Then there exists a unique isomorphism $\varphi : A' \rightarrow A$ such that $\iota_j = \varphi \circ \iota'_j$ for every $j \in J$.

Proof. $(\iota_j : A_j \rightarrow A)_{j \in J}$ satisfy $\iota_{j'} = \iota_j \circ \varphi_{jj'}$ by definition, so they induce a unique morphism $\varphi : A' \rightarrow A$ such that $\iota_j = \varphi \circ \iota'_j$. It remains to construct an inverse morphism. For $i \in I$ choose $j \in J$ with $i \leq j$, and set $\psi_i := \iota'_j \circ \varphi_{ij} : A_i \rightarrow A'$. Note that ψ_i does not depend on the choice of j : Let $j' \in J$ be another element with $i \leq j'$, since J is directed we may assume $j \leq j'$. (Otherwise pick $j'' \in J$ with $j \leq j''$ and $j' \leq j''$ and apply the result to both of these.) Then $\iota'_{j'} \circ \varphi_{ij'} = \iota'_{j'} \circ \varphi_{jj'} \circ \varphi_{ij} = \iota'_j \circ \varphi_{ij}$. Now let $i \leq k \in I$, and choose $J \ni j \geq k$. Then $\psi_k \circ \varphi_{ik} = \iota'_j \circ \varphi_{kj} \circ \varphi_{ik} = \iota'_j \circ \varphi_{ij} = \psi_i$, so the $(\psi_i)_{i \in I}$ induce a morphism $\psi : A \rightarrow A'$ with $\psi_i = \psi \circ \iota_i$.

To show $\psi \circ \varphi = \text{id}_{A'}$, by uniqueness it suffices to show $\psi \circ \varphi \circ \iota'_j = \iota'_j$ for every $j \in J$. Indeed, $\psi \circ \varphi \circ \iota'_j = \psi \circ \iota_j = \psi_j = \iota'_j \circ \varphi_{jj} = \iota'_j$. Similarly, $\varphi \circ \psi \circ \iota_i = \varphi \circ \psi_i = \varphi \circ \iota'_j \circ \varphi_{ij} = \iota_j \circ \varphi_{ij} = \iota_i$ for every $i \in I$, where $i \leq j \in J$, implies $\varphi \circ \psi = \text{id}_A$. $\square$

Back to the exercise: The directed system in question is $(\{U \subseteq X_{\text{open}} \mid x \in U\}, \supseteq)$ so by definition of a base of a topological space it is clear that $(\{U \in \mathcal{B} \mid x \in U\}, \supseteq)$ is a sub-directed set.

(2) From the proposition and (1), we immediately have

$$A_\rho \cong \mathcal{O}_{\text{Spec } A, \rho} = \varinjlim_{U \ni \rho} \mathcal{O}_{\text{Spec } A}(U) \cong \varinjlim_{D(f) \ni \rho} \mathcal{O}_{\text{Spec } A}(D(f)) \cong \varinjlim_{D(f) \ni \rho} A_f.$$

By commutative algebra alone, for $f \notin \rho$, f is invertible in A_ρ by definition, so the universal property yields a unique morphism φ_f s.t. $\pi = \varphi_f \circ \pi_f$, where $\pi : A \rightarrow A_\rho$, $\pi_f : A \rightarrow A_f$ are the canonical maps. Write further $\varphi_{ij} : A_f \rightarrow A_h$ for the maps constructed in exercise 3(2). The $(\varphi_{fh})_{D(f) \supseteq D(h)}$ is clearly a directed system, and $\varphi_h \circ \varphi_{fh}$ satisfies the universal property of φ_f , hence they are equal. Thus they induce a morphism $\varphi : \varinjlim_{D(f) \ni \rho} A_f \rightarrow A_\rho$, given by $[\frac{a}{f^n} \in A_f] \mapsto \frac{a}{f^n} \in A_\rho$. φ is clearly surjective, let $\alpha = [\frac{a}{f^n} \in A_f] \in \ker \varphi$. Then $as = 0$ for some $s \notin \rho$, hence $\frac{a}{f^n} = \frac{as}{f^n s} = 0$ in $A_{f^n s}$, i.e. $\alpha = 0$.

Exercise 1

(1) $f^{-1}(D(h)) = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi^{-1}(\mathfrak{p}) \in D(h)\} = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi(h) \notin \mathfrak{p}\} = D(\varphi(h)).$

(2) Let $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$. Using (1), we have a commutative diagram

$$\begin{array}{ccccccc}
 & & \varphi & & & & \\
 & \swarrow & & \searrow & & & \\
 B & \xleftarrow{\cong} & \mathcal{O}_X(X) & \xleftarrow{f_Y^\#} & \mathcal{O}_Y(Y) & \xleftarrow{\cong} & A \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 B_{\varphi(h)} & \xleftarrow{\cong} & \mathcal{O}_X(D(\varphi(h))) & \xleftarrow{f_{D(h)}^\#} & \mathcal{O}_Y(D(h)) & \xleftarrow{\cong} & A_h \\
 & \nwarrow & & \nearrow & & \nwarrow & \\
 & & \varphi_h & & & &
 \end{array}$$

where the vertical arrows are localizations/restrictions and the isomorphisms on the left and right are proposition 2.11. The diagram of solid arrows commutes: The central square by the definition of a morphism of sheaves, the outer squares by the proof of proposition 2.11, the upper part by proposition 2.18. Hence the composition along the bottom row satisfies the universal property of φ_h , so they agree. In other words, the map $f_{D(h)}^\#$ is, up to identifications of sections with A_h resp. $B_{\varphi(h)}$, exactly the localization morphism φ_h .

Exercise 2

Follows directly from proposition 1.11 and exercise 3.4(1).

Exercise 3

(1) The localization map induces an inclusion-preserving bijection between the prime ideals of $S^{-1}A$ and the prime ideals of A not intersecting S .

(2) $\operatorname{im} f = \{\mathfrak{p} \in \operatorname{Spec} A \mid \varphi(\mathfrak{p}) \in \operatorname{Spec} A_g\} \stackrel{(1)}{=} \{\mathfrak{p} \in \operatorname{Spec} A \mid g \notin \mathfrak{p}\} = D(g)$. In the same way one shows $f(D(h)) = D(h) \cap D(g)$, so f is an open continuous map, i.e. a homeomorphism onto its image.

(3) By (2) we have a map $(f, f^\#) : \operatorname{Spec} A_g \rightarrow (D(g), \mathcal{O}_{\operatorname{Spec} A}|_{D(g)})$, which is a homeomorphism on topological spaces. It remains to check that $f^\#$ is an isomorphism, for which by exercise 2 it suffices to consider principal opens $D(h)$, $D(h) \subseteq D(g)$. Then by exercise 1, $f_{D(h)}^\#$ is canonically identified with the localization map $A_h \rightarrow (A_g)_h$, which is an isomorphism since they satisfy the same universal property.

Exercise 4

By proposition 2.25, $\operatorname{Hom}_{\operatorname{Sch}}(_, \operatorname{Spec} A)$ is a sheaf on X , and since $\operatorname{Hom}_{\operatorname{Ring}}(A, _)$ preserves limits, so is $\operatorname{Hom}_{\operatorname{Ring}}(A, \mathcal{O}_X(_))$. Checking that the given morphism between those sheaves is an isomorphism can be done on a basis by exercise 2, so by lemma 2.22 it suffices to consider X affine, which was done in proposition 2.18.

Exercise 1

(1) $\mathcal{O}_{X,\xi} \cong A_{(0)}$ is the localization at $A \setminus 0$, i.e. the fraction field.

(2) Let U be open. Since $\xi \in U$, by definition of the direct limit, we have morphisms $\varphi_U : \mathcal{O}_X(U) \rightarrow \mathcal{O}_{X,\xi} \rightarrow \text{Frac } A$ such that $\varphi_V \circ \rho_{UV} = \varphi_U$ whenever $V \subseteq U$. This exactly says that the $(\varphi_U)_U$ form a morphism of sheaves $\varphi : \mathcal{O}_X \rightarrow \mathcal{F}$. We can check injectivity on the stalks, so let $\mathfrak{p} \in X$. Then by construction of the structure sheaf, $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}} \rightarrow A_{(0)} \cong \mathcal{F}_{\mathfrak{p}}$ is the map induced by localization, which is injective since A is a domain.

Exercise 2

(1) First suppose X is reduced, let $x \in X$. Since x lies in some affine and the stalk can be computed on it by lemma 1, wlog let $X = \text{Spec } A$ be affine. Now suppose $(\frac{a}{s})^n = 0$ for some $\frac{a}{s} \in A_{\mathfrak{p}}$, $n \geq 1$. Then $s'a^n = 0$ for some $s' \notin \mathfrak{p}$, thus $(s'a)^n = s'^{n-1}(s'a^n) = 0$. Since $\mathcal{O}_X(X) = A$ is reduced by assumption, $s'a = 0$, hence $\frac{a}{s} = 0$ and $A_{\mathfrak{p}}$ is reduced.

Conversely, suppose all stalks $\mathcal{O}_{X,x}$ are reduced, and let $U \subseteq X$ be open. Suppose $s^n = 0$ for some $s \in \mathcal{O}_X(U)$. For every $x \in U$, then $[(U, s^n)] = [(U, s)]^n = 0 \in \mathcal{O}_{X,x}$, so $[(U, s)] = 0$, which means $s|_{V_x} = 0$ for some open $x \in V_x \subseteq U$. But the $(V_x)_{x \in U}$ cover U , so $s = 0$ by the sheaf property.

(2) By the proof of (1), A reduced implies that all stalks of $\text{Spec } A$ are reduced, so we conclude with (1). The converse follows immediately from (1).

Exercise 3

(1) First suppose $\xi = \text{Nil } A$ is prime and let $X = X_1 \cup X_2$ with X_1, X_2 closed. Wlog let $\xi \in X_1$, then $X_1 \supseteq \overline{\{\xi\}} = X$, since the nilradical is contained in every prime ideal. Conversely, suppose X is irreducible and let $p, q \in A$ with $pq \in \text{Nil } A$. Then $V(p) \cup V(q) = V(pq) = V(0) = X$, so wlog $V(p) = X$, i.e. $p \in \text{Nil } A$.

(2) If X is integral, then in particular so is $\mathcal{O}_X(X) = A$. Conversely, if A is integral, then by exercise 1 $\mathcal{O}_X(U) \subseteq \text{Frac } A$ for all $U \subseteq X$ open, and subrings of integral domains are integral domains.

Exercise 4

Let k be a field and consider $X = \text{Spec}(k \times k) = \text{Spec } k \sqcup \text{Spec } k$. Then the stalks are just k , which are integral domains, but $k \times k$ is not.

Exercise 1

Let $\mathfrak{p} \subseteq A$ be a prime ideal. Since $X_i^2 = 0 \in \mathfrak{p}$, we must have $X_i \in \mathfrak{p}$ for all $i \in \mathbb{N}$, hence $\mathfrak{m} := (X_1, X_2, \dots) \subseteq \mathfrak{p}$. But this is already a maximal ideal, since $A/\mathfrak{m} \cong K$ is a field. Thus $\text{Spec } A = \{\mathfrak{m}\}$ is a singleton, and hence in particular Noetherian. But A is not a Noetherian ring due to the strictly increasing chain $(X_1) \subsetneq (X_1, X_2) \subsetneq (X_1, X_2, X_3) \subsetneq \dots$ of ideals in A .

Exercise 2

(1) First let X be Noetherian, $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover. Let U_0 be any of the U_i . We recursively define a sequence $(U_n)_{n \geq 0}$ as follows: If $\bigcup_{j=0}^n U_j = U$, then we have found a finite subcover and are done. Otherwise, there is $x \in U \setminus \bigcup_{j=0}^n U_j$, then let U_{n+1} be some U_i containing x . This procedure cannot proceed indefinitely, since then $(\bigcup_{j=0}^n U_j)_{n \geq 0}$ would be a strictly ascending chain of opens. Conversely, if any open in X is quasi-compact and $U_0 \subseteq U_1 \subseteq \dots$ is an ascending chain of opens, let $U := \bigcup_j U_j$. Since U is quasi-compact by assumption, there exists a finite subcover $U = \bigcup_{k=1}^n U_{j_k}$. Thus the chain stabilizes at $\max(j_1, \dots, j_n)$.

(2) Follows directly from (1) since any open in U is also open in X .

Exercise 3

Lemma 2. Let $\mathcal{F}$ be a presheaf and $\mathcal{G}$ be a sheaf on a topological space X , and let $\mathcal{B}$ be a basis of X . Let $\varphi_B : \mathcal{F}(B) \rightarrow \mathcal{G}(B)$, $B \in \mathcal{B}$ be a collection of morphisms, such that whenever $B' \subseteq B$, $B', B \in \mathcal{B}$, then $\varphi_{B'} \circ \rho_{BB'} = \rho_{BB'} \circ \varphi_B$, where the ρ 's are the restriction maps of $\mathcal{F}, \mathcal{G}$. Then there exists a unique morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ with $\varphi(B) = \varphi_B$ for all $B \in \mathcal{B}$.

Proof. Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} B_i$ be a cover by basis elements. Let $s \in \mathcal{F}(U)$. The elements $\varphi_{B_i}(s)$ glue by assumption to a unique element $\varphi(U)(s) \in \mathcal{G}(U)$, since $B_i \cap B_j$ is again covered by basis elements. We have to check that this definition of $\varphi(U)$ is independent of the given cover, then everything follows immediately.

So let $U = \bigcup_{j \in J} B_j$ be another cover, and denote by $\varphi'(U)$ the associated morphism. Let $B_i \cap B_j = \bigcup_k B_{ijk}$ for all i, j , so that $U = \bigcup_{i,j,k} B_{ijk}$ is a common refinement of the two covers. Then

$$\begin{aligned} \varphi(U)(s)|_{B_{ijk}} &= \varphi(U)(s)|_{B_i}|_{B_{ijk}} = \varphi_{B_i}(s)|_{B_{ijk}} = \varphi_{B_{ijk}}(s) \\ &= \varphi_{B_j}(s)|_{B_{ijk}} = \varphi'(U)(s)|_{B_j}|_{B_{ijk}} = \varphi'(U)(s)|_{B_{ijk}}, \end{aligned}$$

hence $\varphi(U) = \varphi'(U)$ by the sheaf condition. $\square$

(1) Clearly X_{red} is a locally ringed space.

First let $X = \text{Spec } A$ be an affine scheme. We show that $X_{\text{red}} \cong \text{Spec } A_{\text{red}}$: Since $\text{Nil } A$ is the intersection of all prime ideals of A , the topological map $f : \text{Spec } A_{\text{red}} \rightarrow \text{Spec } A$ associated to $A \mapsto A/\text{Nil } A$ is a homeomorphism. Denote by $X_{\text{red}}^{\text{pre}}$ the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$ from the definition of X_{red} . Let $g \in A$ and set $\varphi_{D(g)} : (\mathcal{O}_X)_{\text{red}}^{\text{pre}} = (A_g)_{\text{red}} \rightarrow (A_{\text{red}})_g \cong \mathcal{O}_{\text{Spec } A_{\text{red}}} \cong f_* \mathcal{O}_{\text{Spec } A_{\text{red}}}$ to be the natural isomorphism (localization commutes with quotients). By naturality, this isomorphism commutes with restrictions, so by lemma 2 we obtain a morphism of presheaves $\varphi : (\mathcal{O}_X)_{\text{red}}^{\text{pre}} \rightarrow f_* \mathcal{O}_{\text{Spec } A_{\text{red}}}$. Since its codomain is a sheaf, by the universal property of sheafification this induces a morphism $f^\sharp : (\mathcal{O}_X)_{\text{red}} \rightarrow f_* \mathcal{O}_{\text{Spec } A_{\text{red}}}$ of sheaves. Since all the φ_B were isomorphisms, so are the induced morphisms of stalks $f_x^\sharp = \varphi_x$, $x \in X$. Therefore $(f, f^\sharp) : \text{Spec } A_{\text{red}} \rightarrow X_{\text{red}}$ is an isomorphism of schemes.

Now let X be arbitrary, and $\text{Spec } A \cong U \subseteq X$ be an affine open subset. Then $(\mathcal{O}_X)_{\text{red}}|_U = (\mathcal{O}_U)_{\text{red}} \cong \text{Spec } A_{\text{red}}$, so X is locally isomorphic to affine schemes, i.e. a scheme. Also, it is clear that X is reduced, since that can be checked on stalks, and $(A_{\text{red}})_{\mathfrak{p}} \cong (A_{\mathfrak{p}})_{\text{red}}$ is reduced, for any $\mathfrak{p} \in \text{Spec } A \subseteq X$.

To construct i , take the identity on topological spaces, and the obvious map $\mathcal{O}_X(U) \rightarrow \mathcal{O}_X(U)_{\text{red}}$, followed by the sheafification map. On an affine open subset, this is exactly the map associated to $A \mapsto A_{\text{red}}$.

(2) On topological spaces, take $g = f$, then $f = i \circ g$ is clear. Consider the diagram

$$\begin{array}{ccc} f_*\mathcal{O}_X & \xleftarrow{f^\#} & \mathcal{O}_Y \\ \uparrow g^\# & \swarrow i^\# & \downarrow \text{red} \\ (\mathcal{O}_Y)_{\text{red}} & \xleftarrow{(-)^+} & (\mathcal{O}_Y)_{\text{red}}^{\text{pre}} \end{array}$$

Let $U \subseteq Y$ be open. Since $f_*\mathcal{O}_X(U)$ is reduced, $f_U^\#$ factors through $\mathcal{O}_Y(U)_{\text{red}}$. This gives a morphism of presheaves $h : (\mathcal{O}_Y)_{\text{red}}^{\text{pre}} \rightarrow f_*\mathcal{O}_X$, hence by the universal property of sheafification a morphism $g^\# : (\mathcal{O}_Y)_{\text{red}} \rightarrow f_*\mathcal{O}_X$ with $g^\# \circ i^\# = g^\# \circ (-)^+ \circ \text{red} = h \circ \text{red} = f^\#$ as required.

Exercise 4

(1) Clearly $0 \in I \subseteq I^S$. Let $a, a' \in I^S$, with $sa, s'a' \in I$ for suitable $s, s' \in S$. Then $ss'(a + a') = s'(sa) + s(s'a') \in I$, hence $a + a' \in I^S$. For $r \in A$ we have $s(ra) = r(sa) \in I$, so $ra \in I^S$.

(2) Let $a \in I^S$, $sa \in I$ for $s \in S$. Then $\varphi(a) = \frac{a}{1} = \frac{sa}{s} \in S^{-1}I$. Conversely, let $a \in \varphi^{-1}(S^{-1}I)$, say $\frac{a}{1}\varphi(a) = \frac{i}{s}$. Then there is $s' \in S$ with $s'(as - i) = 0$, i.e. $s'sa = s'i \in I$, so $a \in I^S$.

(3) Let

$$\begin{aligned} \Phi : \{I \subseteq A \text{ ideal} \mid I^S = I\} &\leftrightarrow \{\text{ideals } J \subseteq S^{-1}A\} : \Psi \\ I &\mapsto S^{-1}I \\ \varphi^{-1}(J) &\leftarrow J \end{aligned}$$

Ψ is well-defined since if $a \in (\varphi^{-1}(J))^S$, say $sa \in \varphi^{-1}(J)$, then $\frac{sa}{1} \in J$. Since s is invertible in $S^{-1}A$, it follows that $\frac{a}{1} \in J$ and hence $a \in \varphi^{-1}(J)$.

Now by (2), $\Psi(\Phi(I)) = \varphi^{-1}(S^{-1}I) = I^S$ shows $\Psi \circ \Phi = \text{id}$, so Φ is injective. Let $J \subseteq S^{-1}A$ be an ideal, $\frac{a}{s} \in J$. As above, it follows that $a \in \varphi^{-1}(J)$ and hence $\frac{a}{s} \in S^{-1}(\varphi^{-1}(J))$, so $J \subseteq S^{-1}(\varphi^{-1}(J))$. Conversely, let $\frac{a}{s} \in S^{-1}(\varphi^{-1}(J))$ with $\varphi(a) \in J$. Then $\frac{a}{s} = \frac{1}{s}\varphi(a) \in J$. Therefore, $\Phi \circ \Psi = \text{id}$ as well.

Exercise 1

(1) By a previous exercise, $X \rightarrow \operatorname{Spec} A$ is equivalent to a ring morphism $A \rightarrow \mathcal{O}_X(X)$, and composition with the restriction maps $\mathcal{O}_X(X) \rightarrow \mathcal{O}_X(U)$ gives a unique A -algebra structure on the sheaf $\mathcal{O}_X$.

(2) There are unique maps $\mathbb{Z} \rightarrow \mathcal{O}_X(U)$ for any $U \subseteq X$ open. By uniqueness, they form a sheaf of $\mathbb{Z}$ -algebras, hence by (1) X is a sheaf over $\mathbb{Z}$ in a unique way.

Exercise 2

(1) We apply lemma 3.19 with the property $\mathcal{P}(\operatorname{Spec} B) = B$ is a finitely generated A -algebra. If B is finitely generated and $f \in B$, then $B_f \cong B[X]/(Xf - 1)$ is finitely generated over B , hence also over A . Now let $(b_1, \dots, b_n) = 1$ and assume that all B_{b_i} are finitely generated A -algebras. Write $1 = \sum_{i=1}^n c_i b_i$. For fixed i , let $\frac{a_{i1}}{b_i^{k_1}}, \dots, \frac{a_{im}}{b_i^{k_m}}$ be a generating set, and set $N_i = \{a_{i1}, \dots, a_{im}\}$. Let $S = \bigcup_{i=1}^n N_i \cup \{b_1, \dots, b_n\} \cup \{c_1, \dots, c_n\}$. We show that B is generated by S . Let $\varphi : C = A[S] \hookrightarrow B$ be the inclusion; then $\varphi_{b_i} : C_{b_i} \hookrightarrow B_{b_i}$ is surjective, since C_{b_i} contains $\frac{1}{b_i} N_i$. Hence $C_{b_i} \cong B_{b_i}$ for all b_i .

Now a standard local-global argument shows $C = B$: Let $M = B/C$, then $M_{b_i} \cong B_{b_i}/C_{b_i} = 0$. Let $m \in M$. Since $\frac{m}{1} = 0 \in M_{b_i}$, there exists $k \in \mathbb{N}$ with $b_i^k m = 0$ for all $i = 1, \dots, n$. Therefore $(m) = (m)(b_1^k, \dots, b_n^k) = (b_1^k m, \dots, b_n^k m) = 0$, hence $M = 0$ and $B = C$ is finitely generated as an A -algebra.

(2) Clear.

Exercise 3

Let $(i, i^\#) : \operatorname{Spec} K \rightarrow X$ be a morphism of schemes. Since $\operatorname{Spec} K$ has one point, $i(\operatorname{Spec} K) = \{x\}$ is a singleton. Hence $i_* \mathcal{O}_{\operatorname{Spec} K}$ is the skyscraper sheaf with value K at x , so the map $i^\# : \mathcal{O}_X \rightarrow i_* \mathcal{O}_{\operatorname{Spec} K}$ is uniquely determined by its induced map of stalks $i_x : \mathcal{O}_{X,x} \cong (i_* \mathcal{O}_{\operatorname{Spec} K})_x = K$, since we must have $i_U^\#(s) = i_x([U, s])$ for any $U \subseteq X$ open and $s \in \mathcal{O}_X(U)$. Now i_x is local by definition, so $i_x(\mathfrak{m}_x) = 0$, so i_x factors through $\alpha : k(x) \rightarrow K$. Therefore we have constructed a map from $X(K)$ to pairs (x, α) as desired. Conversely, given (x, α) , set $i((0)) = x$ and $i_U^\#(s) = \alpha(\bar{s})$ for all $U \subseteq X$ open and $s \in \mathcal{O}_X(U)$. This clearly gives an element of $\operatorname{Spec} K \rightarrow X$. These two constructions are obviously inverse to each other.

Exercise 4

(1) The structure morphism $X \rightarrow \operatorname{Spec} k$ induces on stalks a field extension $k \rightarrow \mathcal{O}_{X,x} \rightarrow k(x)$. By exercise (3), if $(x, \alpha) \in X(K)$, then $k \rightarrow k(x) \rightarrow k$ is the identity, so both maps have to be isomorphisms. The converse is clear.

(2) Clear from (1), since $\mathcal{O}_{X,x} \cong \mathcal{O}_{U,x}$.

(3) By proposition 2.18, we are looking for morphisms $\varphi : A = k[X_1, \dots, X_n]/(f_1, \dots, f_m) \rightarrow k$ such that precomposition with $i : k \hookrightarrow A$ is the identity. By the universal properties of polynomial rings and quotients, those maps φ correspond exactly to tuples $x \in k^n$ with $f_i(x) = 0$ for all i .

(4) By (3), we are looking for real solutions of $0 = X^2 + Y^2 + 1 \geq 1$, so $X(\mathbb{R}) = \emptyset$. Let $x = (X, Y^2 + 1) \in X$. Then $(\mathbb{R}[X, Y]/(X^2 + Y^2 + 1))/x \cong \mathbb{R}[Y]/(Y^2 + 1) \cong \mathbb{C}$ as $\mathbb{R}$ -algebras via $Y \mapsto \pm i$, so we get two morphism $i : \operatorname{Spec} \mathbb{C} \rightarrow X$ of $\mathbb{R}$ -schemes.

Exercise 1

First let x be a closed point, with corresponding maximal ideal $\mathfrak{m}$. Then $A/\mathfrak{m}$ is already a field, so $k(x) = A/\mathfrak{m}$, which is finitely generated by $\{\bar{1}\}$, hence algebraic by Zariski's Lemma. Conversely, suppose $K = \text{Frac}(A/\mathfrak{p})/k$ is algebraic. Then $A/\mathfrak{p}$ is finitely generated over k , and every generator is algebraic, hence $A/\mathfrak{p}$ is finite over k . But every integral domain finite over a field is itself a field, since then multiplication by any nonzero element is injective, hence surjective.

Exercise 2

We have to prove that every open contains a closed point. It clearly suffices to do this for a basis, so let $\emptyset \neq \text{Spec } A \subseteq X$. Since $A \neq 0$, $\text{Spec } A$ has a maximal ideal/closed point x . By exercise 1, this remains a closed point in X , since $k(x)$ does not depend on the neighbourhood of x .

Exercise 3

$\text{Spec } k[[t]] = \{0, (t)\}$ and hence $\{0\}$ is an open subset without a closed point. Thus by exercise 2, X is not locally of finite type over k , i.e. not a finitely generated k -algebra by sheet 7, exercise 2.

Exercise 4

(1) Write $d := \deg f$. Let $\mathfrak{p} \in D_+(f)$ and $x \in \mathfrak{p}$ homogeneous of degree n . Then $\frac{x^d}{f^n} \in \mathfrak{p}B_f \cap B_{(f)} = \theta(\mathfrak{p})$, hence by definition $x \in \psi(\theta(\mathfrak{p}))_n \subseteq \psi(\theta(\mathfrak{p}))$ by definition. Since $\mathfrak{p}$ is homogeneous, this implies $\mathfrak{p} \subseteq \psi(\theta(\mathfrak{p}))$. Conversely, let $x \in \psi(\theta(\mathfrak{p}))_n$ for some n , i.e. $\frac{x^d}{f^n} \in \theta(\mathfrak{p})$. This implies $f^i x^d \in \mathfrak{p}$ for some i , hence $x \in \mathfrak{p}$ since $\mathfrak{p}$ is prime. Again by homogeneity, $\psi(\theta(\mathfrak{p})) \subseteq \mathfrak{p}$ and hence $\psi \circ \theta = \text{id}_{D_+(f)}$.

Conversely, let $\mathfrak{q} \in \text{Spec } B_{(f)}$ and $\frac{x}{f^n} \in \mathfrak{q}$. Then $x^d \in \psi(\mathfrak{q})_{dn} \subseteq \psi(\mathfrak{q})$, so $x \in \psi(\mathfrak{q})$ and hence $\frac{x}{f^n} \in \theta(\psi(\mathfrak{q}))$. Conversely, let $\frac{x}{f^n} \in \theta(\psi(\mathfrak{q}))$. Then $f^i x \in \psi(\mathfrak{q})$ for some i , and since $\psi(\mathfrak{q}) \in D_+(f)$ even $x \in \psi(\mathfrak{q})_{dn}$. By definition, this means $(\frac{x}{f^n})^d = \frac{x^d}{f^{dn}} \in \mathfrak{q}$, i.e. $\frac{x}{f^n} \in \mathfrak{q}$. This shows $\theta \circ \psi = \text{id}_{\text{Spec } B_{(f)}}$.

(2) Let $\mathfrak{p} \in \text{Proj } B$. Then $h \in \theta(\mathfrak{p})$ if and only if $g \in \psi(\theta(\mathfrak{p}))_{\deg g} \subseteq \psi(\theta(\mathfrak{p})) = \mathfrak{p}$ by definition, so $\theta(D_+(g)) = D(h)$.